Full Stack Web Development

Innomatics Research Labs — Course Syllabus

The official Innomatics Full Stack page presents Java Full Stack, Python Full Stack and MERN Full Stack tracks covering frontend, backend, databases, APIs, DSA/development practice and projects.

Source: https://www.innomatics.in/full-stack-web-development-course-with-job-assistance/

Course Structure: Full Stack Development

- The official page presents three learning tracks: Java Full Stack, Python Full Stack and MERN Full Stack.
- All tracks include frontend, backend, database and project-oriented development.

Java Full Stack — Module 1: Coding with VS Code & Version Control

- Programming basics and SDLC
- VS Code setup and workspace management
- Command line and Git fundamentals
- GitHub: init, add, commit, push, pull, clone, branching and merging
- First project execution

Java Full Stack — Module 2: HTML, CSS & Bootstrap

- HTML elements, forms, media, links and utility tags
- CSS selectors, colors, backgrounds and box model
- Flexbox, Grid, positioning, transformations and animations
- Bootstrap typography, buttons, forms, tables, navigation and responsive grid

Java Full Stack — Module 3: JavaScript, AJAX & API Integration

- ES6 variables, data types, operators and control flow
- Functions, arrow functions, classes and OOP
- DOM, events and higher-order functions
- Client-server architecture, XML/JSON and APIs
- Postman, asynchronous programming and frontend API integration

Java Full Stack — Module 4: React.js

- React setup, folder structure and components
- JSX, events, props and state
- Forms and inputs
- Hooks: useState and useEffect
- Axios, routing and Context API
- Redux and middleware
- Third-party APIs
- Frontend capstone and deployment readiness

Java Full Stack — Module 5: Core & Intermediate Java

- Java setup and program structure

- Control flow, arrays, loops and methods
- Classes and objects
- Inheritance, polymorphism, encapsulation and abstraction
- Collections: List, Set, Map, ArrayList, LinkedList
- Streams and lambda expressions

Java Full Stack — Module 6: Advanced Java & Real-World Concepts

- Exception handling and exception hierarchy
- Checked vs unchecked exceptions
- Custom exceptions
- File handling
- Backend-oriented Java concepts

Java Full Stack — Module 7: SQL, JDBC & Databases

- MySQL fundamentals and RDBMS concepts
- CRUD, filtering, grouping and joins
- Subqueries and complex queries
- JDBC architecture and drivers
- Connections, statements, PreparedStatement and ResultSet
- Secure database coding practices
- HTTP, web applications and MVC concepts
- Servlets, JSP and servlet containers

Java Full Stack — Module 8: Spring Boot

- Spring Boot overview and setup
- Dependency Injection and Inversion of Control
- Building backend applications with Spring Boot

Python Full Stack — Module 1: Coding with VS Code & Version Control

- Programming basics, SDLC, VS Code, file management, command line, Git and GitHub

Python Full Stack — Module 2: HTML, CSS & Bootstrap

- HTML, forms, media and links
- CSS selectors, box model, Flexbox, Grid and responsive design
- Bootstrap UI components and grid system

Python Full Stack — Module 3: JavaScript, AJAX & APIs

- JavaScript ES6, functions, OOP, DOM and events
- Higher-order functions
- AJAX, JSON/XML, Postman, async programming and API integration

Python Full Stack — Module 4: React.js

- React components, JSX, events, props/state, forms and hooks
- Axios, routing, Context API, Redux and third-party APIs

- Frontend capstone

Python Full Stack — Module 5: Python Programming Foundations & Intermediate

- Python foundations
- OOP: inheritance, polymorphism, encapsulation and abstraction
- Comprehensions, iterators, generators and lambda
- map, filter and reduce

Python Full Stack — Module 6: Advanced Python Concepts

- Advanced Python concepts
- File handling: read/write/append, modes, context managers and data-processing use cases

Python Full Stack — Module 7: MySQL/MongoDB Databases

- MySQL/PostgreSQL fundamentals, CRUD, filtering, grouping and joins
- Python DB-API connectivity
- ORM concepts and SQLAlchemy
- CRUD using ORM and secure database interaction

Python Full Stack — Module 8: Django & Flask

- HTTP and client-server architecture
- Django MVT architecture and project structure
- Django setup, settings, URLs, views and templates
- Middleware, forms and basic REST APIs
- Django ORM, relationships and migrations
- Authentication, authorization, sessions and cookies
- Flask-based backend concepts

MERN Full Stack — Module 1: Coding & Version Control

- Programming basics, SDLC, VS Code, command line, Git and GitHub

MERN Full Stack — Module 2: HTML, CSS & Bootstrap

- HTML, CSS, layout, responsive design and Bootstrap

MERN Full Stack — Module 3: JavaScript, AJAX & API Integration

- ES6, functions, OOP, DOM, events, higher-order functions
- Client-server model, APIs, Postman and asynchronous programming

MERN Full Stack — Module 4: React.js

- React setup, components, JSX, props/state, hooks, Axios, routing, Context API and Redux
- Frontend capstone and API integration

MERN Full Stack — Module 5: Core Java Programming

- Java environment, program structure, control flow, arrays, loops and methods
- Classes/objects

- OOP principles, collections, streams and lambdas

MERN Full Stack — Module 6: MERN Backend with Node.js

- Node.js and V8
- Node vs browser JavaScript
- Event loop and non-blocking model
- Globals, environment variables and CLI arguments
- CommonJS/ES Modules, npm and package.json
- Core modules, file system, HTTP server and routing
- Callbacks, promises and async/await
- Events, streams, concurrency and scalability
- Mini file-based API project

MERN Full Stack — Module 7: Express.js

- Express setup, routing and request/response
- Middleware and custom middleware
- MVC structure and services layer
- Validation, sanitization and error handling
- REST principles and CRUD APIs
- Pagination, filtering, sorting and search
- Full API project with authentication and CRUD

MERN Full Stack — Module 8: MongoDB

- SQL vs NoSQL
- MongoDB architecture, collections, documents and BSON
- MongoDB setup, databases and collections
- MongoDB-oriented backend development

Full Stack Projects & Practical Work

- Frontend capstone applications
- API integration projects
- State-management implementations
- Backend/API projects and deployment-oriented work
- Hands-on assignments and real-world development practice

Prepared from the syllabus/curriculum published on the official Innomatics Research Labs website. Website content can change; use the source URL above as the authoritative current reference.